HW #7

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. (Do not use §4.3-4.5!)

(1) Let A be a Dedekind domain. Show that there is a one-to-one correspondence

$$\{\text{Nonzero prime ideals of } A\} \leftrightarrow \left\{ \begin{array}{c} \text{Normalized discrete valuations } v \text{ on} \\ \mathrm{Frac}(A) \text{ such that } v(x) \geq 0 \text{ for all } x \in A \end{array} \right\}$$

$$\mathfrak{p} \subset A \mapsto v_{\mathfrak{p}} \text{ induced by the DVR } A_{\mathfrak{p}}$$

$$\{x \in A \ : \ v(x) \geq 1\} \leftarrow\!\shortmid v.$$

(2) Let L/K be an extension of number fields, and let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. Show that the one-to-one correspondence of (1) for $A = \mathcal{O}_L$ restricts to a one-to-one correspondence

$$\{\text{Prime ideals } \mathfrak{q} \text{ of } \mathcal{O}_L \text{ dividing } \mathfrak{p}\} \leftrightarrow \left\{ \begin{array}{c} \text{Normalized discrete valuations } v \text{ on } L \text{ such that} \\ v|_K = ev_{\mathfrak{p}} \text{ for some } e \in \mathbb{Z}_{\geq 1} \end{array} \right\}.$$

Here, $v|_K$ is the restriction of $v : L \to \mathbb{Z}_{\geq 0} \cup \{\infty\}$ to K (i.e., the composition of v with the inclusion $K \hookrightarrow L$).

(3) In the situation of (2), if $\mathfrak{q}$ is a prime ideal of $\mathcal{O}_L$ dividing $\mathfrak{p}$, show that $v_{\mathfrak{q}} = e(\mathfrak{q}|\mathfrak{p})v_{\mathfrak{p}}$.

Proof.

(1) We first check that the maps in both ways are well-defined.
- Given a normalized discrete valuation v on $\mathrm{Frac}(A)$, it is clear that $\{x \in A \ : \ v(x) \geq 1\}$ is a prime ideal of A, by the condition that $v(x) \geq 0$ for all $x \in A$.
- If $\mathfrak{p} \subset A$ is a nonzero prime ideal, $v_{\mathfrak{p}}$ is nonnegative on $A_{\mathfrak{p}}$, which contains A.

We check that the two maps are inverses to each other.

Given $\mathfrak{p} \subset A$, $\{x \in \mathrm{Frac}(A) \ : \ v_{\mathfrak{p}}(x) \geq 1\} = \mathfrak{p}A_{\mathfrak{p}}$, so $\{x \in A \ : \ v_{\mathfrak{p}}(x) \geq 1\} = \mathfrak{p}A_{\mathfrak{p}} \cap A = \mathfrak{p}$ (by, for example, Theorem A.40(4)).

Given a normalized discrete valuation v, let $\mathfrak{p} = \{x \in A \ : \ v(x) \geq 1\}$. We want to show that $v = v_{\mathfrak{p}}$. We know that $v \equiv 0$ on $A - \mathfrak{p}$. Therefore, $v \geq 0$ on $(A - \mathfrak{p})^{-1}A = A_{\mathfrak{p}}$. It suffices to show that $v = v_{\mathfrak{p}}$ on $A_{\mathfrak{p}}$. As $A_{\mathfrak{p}}$ is a DVR, there is a uniformizer $\pi \in A_{\mathfrak{p}}$ such that $\mathfrak{p}A_{\mathfrak{p}} = \pi A_{\mathfrak{p}}$. As $v = v_{\mathfrak{p}} = 0$ on $A_{\mathfrak{p}}^{\times}$, to show that $v = v_{\mathfrak{p}}$ on $A_{\mathfrak{p}}$, it suffices to show that $v(\pi) = v_{\mathfrak{p}}(\pi)$. Note that $v_{\mathfrak{p}}(\pi) = 1$. On the other hand, $v(\pi) \geq 1$ by definition of $\mathfrak{p}$. If $v(\pi) > 1$, then any element in $\mathfrak{p}$ would have the value of v greater than 1, which would not make v normalized. Thus, $v(\pi)$ has to be equal to 1, as desired.

(2) If $\mathfrak{q} \subset \mathcal{O}_L$ divides $\mathfrak{p}$, so that $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$, then $v_{\mathfrak{q}}|_K$ will define a discrete valuation that is > 0 on $\mathfrak{p}$. Let v be a normalized valuation on K such that $v_{\mathfrak{q}}|_K = ev$ for some $e \in \mathbb{Z}_{\geq 1}$. Then, v is > 0 on $\mathfrak{p}$, so by (1), it has to be that $v = v_{\mathfrak{p}}$. So the map in $\mapsto$ direction in (1) sends the left hand side to the right hand side.

Conversely, if v is a normalized discrete valuation on L such that $v|_K = ev_{\mathfrak{p}}$ for some $e \in \mathbb{Z}_{\geq 1}$, then $v = v_{\mathfrak{q}}$ for some $\mathfrak{q} \subset \mathcal{O}_L$, such that $v_{\mathfrak{q}}(x) > 0$ for $x \in \mathfrak{p}$. Note that $v_{\mathfrak{q}}(x) > 0$

for $x \in \mathcal{O}_L$ if and only if $x \in \mathfrak{q}$, so this implies that $\mathfrak{p} \subset \mathfrak{q}$, so $\mathfrak{q}$ divides $\mathfrak{p}$. Therefore, the map in $\hookleftarrow$ direction in (1) sends the right hand side to the left hand side.

(3) Clear from Theorem 4.1(2).

$\square$

Question 2. Let K be a number field, and let $\alpha \in K^\times$ that is not a square in K. Consider $L = K(\sqrt{\alpha})$, which is a quadratic extension of K.

(1) Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. If $v_\mathfrak{p}(\alpha)$ is odd, show that $\mathfrak{p}$ is ramified in L.

Hint. Use Question 1.

(2) Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$ that does not divide 2. If $v_\mathfrak{p}(\alpha)$ is even, show that $\mathfrak{p}$ is unramified in L.

Hint. Take a uniformizer of $\mathcal{O}_{K,\mathfrak{p}}$ to reduce to the case of $v_\mathfrak{p}(\alpha) = 0$, and use the relative discriminant.

(3) Find an example of K, α, and $\mathfrak{p}$ (which necessarily divides 2), such that $\mathfrak{p}$ is ramified in L, even though $v_\mathfrak{p}(\alpha)$ is even.

(4) Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$ that divides 2. If α is a square modulo $\mathfrak{p}^{2e(\mathfrak{p}|2)}$ (namely, if there exists $x \in \mathcal{O}_K$ such that $x^2 \equiv \alpha \pmod{\mathfrak{p}^{2e(\mathfrak{p}|2)}}$), show that $\mathfrak{p}$ is unramified in L.

Hint. Take a uniformizer of $\mathcal{O}_{K,\mathfrak{p}}$ to reduce to the case of $v_\mathfrak{p}(\alpha) = 0$. For x as in the problem, show that $\frac{x+\sqrt{\alpha}}{2} \in \mathcal{O}_{K(\sqrt{\alpha}),\mathfrak{p}}$. You need to use that $v_\mathfrak{p}(4) = 2e(\mathfrak{p}|2)$. Use Theorem 4.10 to show that $(\mathcal{O}_K - \mathfrak{p})^{-1} \operatorname{disc}(L/K)$ is the unit ideal of $\mathcal{O}_{K,\mathfrak{p}}$.

Proof.

(1) If $\mathfrak{p}$ is unramified in L, there must be a normalized discrete valuation v on L such that $v|_K = v_\mathfrak{p}$. On the other hand, $v(\alpha) = 2v(\sqrt{\alpha})$ is even, whereas $v(\alpha) = v_\mathfrak{p}(\alpha)$ is odd, a contradiction. Therefore, $\mathfrak{p}$ must be ramified in L.

(2) Take a uniformizer $\pi \in \mathcal{O}_{K,\mathfrak{p}}$ and take $\beta = \frac{\alpha}{\pi^{v_\mathfrak{p}(\alpha)}}$. As $v_\mathfrak{p}(\alpha)$ is even, $K(\sqrt{\alpha}) = K(\sqrt{\beta})$. Thus we can assume without loss of generality that $v_\mathfrak{p}(\alpha) = 0$. Now $\operatorname{disc}(L/K)$ contains $D(1, \sqrt{\alpha}) = \det \begin{pmatrix} \operatorname{Tr}_{L/K}(1) & \operatorname{Tr}_{L/K}(\sqrt{\alpha}) \\ \operatorname{Tr}_{L/K}(\sqrt{\alpha}) & \operatorname{Tr}_{L/K}(\alpha) \end{pmatrix} = \det \begin{pmatrix} 2 & 0 \\ 0 & 2\alpha \end{pmatrix} = 4\alpha$, so in particular $\mathfrak{p}$ does not divide $\operatorname{disc}(L/K)$. Therefore, $\mathfrak{p}$ is unramified in L.

(3) Take $K = \mathbb{Q}$, $\alpha = -1$, and $\mathfrak{p} = (2)$. Even though $v_2(-1) = 0$, 2 is ramified in $\mathbb{Q}(\sqrt{-1})$, as $(2) = (1 + \sqrt{-1})^2$.

(4) Again we can assume in the same way as in (2) that $v_\mathfrak{p}(\alpha) = 0$. Let $x \in \mathcal{O}_K$ be such that $x^2 \equiv \alpha \pmod{\mathfrak{p}^{2e(\mathfrak{p}|2)}}$. Let $\beta = \frac{x+\sqrt{\alpha}}{2} \in L$. The minimal polynomial $p_\beta(X) \in K[X]$ of β over K is

$$p_\beta(X) = X^2 - xX + \frac{x^2 - \alpha}{4}.$$

Note that $x^2 - \alpha \in \mathfrak{p}^{2e(\mathfrak{p}|2)}$. Let $2\mathcal{O}_K = \prod_{i=1}^g \mathfrak{p}_i^{e(\mathfrak{p}_i|2)}$ with $\mathfrak{p}_1 = \mathfrak{p}$. Then, $4\mathcal{O}_K = \prod_{i=1}^g \mathfrak{p}_i^{2e(\mathfrak{p}_i|2)}$. Therefore, $v_{\mathfrak{p}_i}(4) = 2e(\mathfrak{p}_i|2)$. This implies that $v_\mathfrak{p}\left(\frac{x^2-\alpha}{4}\right) \geq 0$, so $\frac{x^2-\alpha}{4} \in \mathcal{O}_{K,\mathfrak{p}}$. This implies that $\beta \in \mathcal{O}_{L,\mathfrak{p}}$. This implies that $(\mathcal{O}_K - \mathfrak{p})^{-1} \operatorname{disc}(L/K)$ contains

$D(1,\beta) = \det\begin{pmatrix} 2 & x \\ x & \frac{x^2+\alpha}{2} \end{pmatrix} = \alpha$, which implies that $(\mathcal{O}_K - \mathfrak{p})^{-1}\operatorname{disc}(L/K)$ is the unit ideal of $\mathcal{O}_{K,\mathfrak{p}}$, so that $\operatorname{disc}(L/K)$ is not divisible by $\mathfrak{p}$, or that $\mathfrak{p}$ is unramified in L.

□

Question 3. Later in the course, when we learn class field theory, we will see that, for any number field K, there is a very special abelian extension H_K/K, called the **Hilbert class field of** K, such that $\operatorname{Gal}(H_K/K) \cong \operatorname{Cl}(K)$ and every prime ideal of K is unramified in H_K.

Although the above general fact is currently beyond our reach, in this question, we will show a very particular consequence of this general fact by hand: **for an imaginary quadratic field K with h_K even, there exists a quadratic extension L/K in which every prime ideal of K is unramified**.

Let $K = \mathbb{Q}(\sqrt{-n})$ be an imaginary quadratic field, with n a squarefree positive integer, such that h_K is even.

(1) Show that $\operatorname{disc}(K)$ has at least two prime factors.

Hint. Use Exercise 3.4.

(2) Use Question 2 to show that, if p is any odd prime divisor of n, then either $K(\sqrt{p})$ or $K(\sqrt{-p})$ is a desired quadratic extension (i.e., where all prime ideals of K are unramified).

Hint. For prime ideals of K not dividing 2, use that fact that all rational prime factors of n are ramified in K. For prime ideals of K dividing 2, use the fact that either p or $-p$ is $1 \pmod 4$. Remember that such a quadratic extension exists if and only if h_K is even, so your proof must use this (e.g., Exercise 3.4) at some point.

Proof.

(1) Exercise 3.4 shows that n is either a composite number or a prime congruent to 1 mod 4. For the former case, $\operatorname{disc}(K)$ is a composite number as n already is. For the latter case, $\operatorname{disc}(K) = -4n$, so again it has two prime factors (2 and n).

(2) For $\alpha = \pm p$, we claim that $K(\sqrt{\alpha})/K$ is always unramified away from 2. By Question 2(2), this will follow if $v_{\mathfrak{p}}(\pm p)$ is even for every prime ideal of $\mathcal{O}_K$ not dividing 2. If $\mathfrak{p}$ does not divide p, then obviously $v_{\mathfrak{p}}(\pm p) = 0$ is even. If $\mathfrak{p}|p$, then as p is ramified, $(p) = \mathfrak{p}^2$. Therefore, $v_{\mathfrak{p}}(\pm p) = 2v_p(\pm p) = 2$ is even as well.

Now let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$ dividing 2. Take α to be p or $-p$ such that $\alpha \equiv 1 \pmod 4$. Let $L = K(\sqrt{\alpha})$. Note that L/K is always a quadratic extension. This is obvious when $n \neq p$, and if $n = p$, then by Exercise 3.4, h_K even implies that $p \equiv 1 \pmod 4$, which implies that $\alpha = p$, which is not a square in $K = \mathbb{Q}(\sqrt{-p})$.

Now by definition α is a square mod $4\mathcal{O}_K$. As $\mathfrak{p}^{2e(\mathfrak{p}|2)}$ divides $4\mathcal{O}_K$, by Question 2(4), $\mathfrak{p}$ is unramified in $K(\sqrt{\alpha})$, as desired.

□